

Affine-past curvature, nonlinear conditional operator, and multi-root globalization boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope: result and stopping boundary

R-151 proves a strict local Hessian gap for the complete zero-past linear $p : 2p$ endpoint. It deliberately leaves three lifts open: nonzero strict past, nonlinear predictable feedback, and owner-complete aggregation over interacting roots. This note performs the promised bounded pass on all three without introducing separately fabricated forest, balanced, trace, or low reserves.

The outcome has one positive theorem and two exact method boundaries.

1. A fixed past field mean does not weaken the R-151 covariance-Hessian bound when the past current mean is zero. An exact stronger Sturm certificate gives endpoint loss below 19/25, hence augmented curvature above 7/50. For nonzero past current mean, the three new signed terms are displayed exactly. A quantitative once-declared current collar preserves the R-151 1/10 margin.
2. For a legal nonlinear fixed-law direction $\phi(\xi_1)$, the exact criterion is an almost-sure conditional source-space operator lower bound. The averaged linear tests of R-151 do not imply it. A smooth translated-bump fixture proves the non-implication, but is not an A1 production counterexample.
3. Pairwise local gaps do not aggregate merely because source coordinates are used once. The full recombined multi-root Hessian must pass one generalized-eigenvalue or kernel-Schur test. A rational two-edge fixture has positive local gaps above 7/50 and a negative global eigenvalue.

Thus the bounded lift does not reveal an owner-complete global mechanism under the registered hypotheses. A13 remains valid conditional mathematics at T4; T-050, Nelson, and Sector A remain open. This is the predeclared stopping boundary for adding further local pair lemmas without the actual production global operator.

2. Exact affine-past Hessian

Condition on the strict-past sigma-field $\mathcal{F}_-$. For almost every past realization let $\mathbb{E}_f = \mathbb{E}[\cdot \mid \mathcal{F}_-]$. In the retained R-151 antipodal chart write

$$W_t = m + X + tZ, \quad V_{t,i} = n_i + Y_i + tU_i, \quad (2.1)$$

where

$$Z = S_2 H \xi_1, \quad U_i = D_{2,i} H \xi_1. \quad (2.2)$$

Here (X, Y) is the centered fresh Gaussian field-current pair, $m = \mathbb{E}[W_0 \mid \mathcal{F}_-]$ and $n_i = \mathbb{E}[V_{0,i} \mid \mathcal{F}_-]$ are fixed after conditioning, and the centered same-point field-current covariance is zero. The centered future

covariance and common-even regulator hypotheses are exactly those of R-151. Let $B = C_6^* C_6$ and let Q_0 be the A7 primitive current covariance. The complete covariance-normal endpoint is

$$\mathcal{R}(t) = \frac{1}{2} \int \mathbb{E}_f \left[\sum_i V_{t,i}^* B(W_t) V_{t,i} - \text{Tr}\{B(W_t) Q_0\} \right] dx. \quad (2.3)$$

Direct differentiation, before any score transfer, gives

$$\mathcal{R}''(0) = \int \mathbb{E}_f \left[\sum_i \{ U_i^* B U_i + 2 U_i^* D B[Z] (Y_i + n_i) + \frac{1}{2} (Y_i + n_i)^* D^2 B[Z, Z] (Y_i + n_i) \} - \frac{1}{2} \text{Tr}(D^2 B[Z, Z] Q_0) \right] dx. \quad (2.4)$$

Separating the centered core from the current mean yields

$$\mathcal{R}''(0) = \mathcal{H}_{\text{core}}(m) + \Delta_n, \quad (2.5)$$

where $\mathcal{H}_{\text{core}}(m)$ is the R-151 covariance Hessian with the field expectation shifted by m , and

$$\Delta_n = \int \mathbb{E}_f \sum_i \{ 2 U_i^* D B[Z] n_i + n_i^* D^2 B[Z, Z] Y_i + \frac{1}{2} n_i^* D^2 B[Z, Z] n_i \} dx. \quad (2.6)$$

The coefficients 2, 1, 1/2 in (2.6) are forced by the derivative of the one scalar endpoint; none is an independently assignable owner.

Theorem 2.1 (arbitrary field mean with zero current mean)

Assume the R-151 retained-lattice chart and a fixed conditional field mean $m(x)$, but $n = 0$. Then, for every nonzero H ,

$$\mathcal{H}_{\text{core}}(m)[H, H] > -\frac{19}{25} \|H\|_{\mathbb{F}}^2. \quad (2.7)$$

The exact source Hessian is $9\|H\|_{\mathbb{F}}^2/10$, and the terminal sixth-power Hessian is nonnegative. Consequently

$$\mathcal{A}''(0)[H, H] > \frac{7}{50} \|H\|_{\mathbb{F}}^2. \quad (2.8)$$

Proof. The R-130 bounds for DB and D^2B are global in the field value. Shifting the Gaussian expectation by an arbitrary fixed m therefore does not change the two signed response constants in the R-151 proof. The positive $\text{Tr}(Q_2 \mathbb{E} B)$ term remains positive. Section 3 proves the stronger radial loss bound 19/25. Adding the source Hessian and the nonnegative sixth-power Hessian gives $9/10 - 19/25 = 7/50$. $\square$

This theorem is a curvature statement. A nonstationary m generally destroys the R-151 Fourier proof that $\mathcal{R}'(0) = 0$, so (2.8) alone does not make the origin a local minimum.

The theorem is proved on the enlarged fixed-mean field/current class in which m and n are algebraically separated. In the production synthesis one normally has $n_i = \partial_i m$; hence production $n = 0$ normally restricts m to a spatially constant torus mode. The enlarged theorem is valid, but does not assert that every such pair (m, n) is production-realizable.

Theorem 2.2 (conditional past-current collar)

Pathwise in the strict past, put

$$\begin{aligned} N &= \operatorname{ess\,sup}_x \left(\sum_i |n_i(x)|^2 \right)^{1/2}, \\ M_W &= \operatorname{ess\,sup}_x \{ \mathbb{E}_f |m + X|^2 \}^{1/2}, \\ M_Y &= \operatorname{ess\,sup}_x \left\{ \mathbb{E}_f \sum_i |Y_i|^2 \right\}^{1/2}. \end{aligned} \quad (2.9)$$

Let $\lambda_2 = \lambda_{\min} A(2p)$ and use the exact R-130 constants

$$L_6 = \frac{1143}{250P}, \quad H_6 = \frac{7083}{500P}, \quad P = 4 + 10^{-12}. \quad (2.10)$$

Define

$$C_{\text{past}} = \frac{8\sqrt{3}L_6|p|}{\lambda_2} NM_W + \frac{4\sqrt{3}H_6}{\lambda_2} NM_Y + \frac{2H_6}{\lambda_2} N^2. \quad (2.11)$$

Then

$$\mathcal{A}''(0)[H, H] > \left(\frac{7}{50} - C_{\text{past}} \right) \|H\|_{\mathbb{F}}^2. \quad (2.12)$$

In particular,

$$C_{\text{past}} \leq \frac{1}{25} \implies \mathcal{A}''(0)[H, H] > \frac{1}{10} \|H\|_{\mathbb{F}}^2. \quad (2.13)$$

Proof. In the spatial-gradient direct sum, the R-130 derivative envelopes and Cauchy-Schwarz give

$$|T_1| \leq 2\sqrt{3}L_6NM_Wbe\|H\|_{\mathbb{F}}^2, \quad |T_2| \leq 2\sqrt{3}H_6NM_Yb^2\|H\|_{\mathbb{F}}^2, \quad |T_3| \leq H_6N^2b^2\|H\|_{\mathbb{F}}^2, \quad (2.14)$$

where $b^2 \leq 2/(V\lambda_2)$ and $e = 2|p|b$. The factor $\sqrt{3}$ is the joint-Gaussian fourth-moment bound $\mathbb{E}|Z|^2|U|^2 \leq 3\mathbb{E}|Z|^2\mathbb{E}|U|^2$ and $\mathbb{E}|Z|^4 \leq 3(\mathbb{E}|Z|^2)^2$. Integrating over volume V gives (2.11). Combine it with (2.8). The implication (2.13) assumes $C_{\text{past}}(\omega) \leq 1/25$ for almost every past realization. If H is past-dependent, an averaged collar alone is insufficient; the corresponding weighted form estimate must control $\mathbb{E}[C_{\text{past}}\|H\|_{\mathbb{F}}^2]$. $\square$

The registered hypotheses supply no uniform bounds on N, M_W, M_Y . Thus (2.13) is a real sufficient theorem, but not an unconditional nonzero-past production theorem. The missing input is either this collar or a once-owned source/sextic estimate that pays exactly Δ_n .

3. Exact radial and momentum certificates

The strict A1 mass floor gives

$$f(x) = x^2 - \frac{4626377063}{5000000000}x + \frac{5020336473}{10000000000}. \quad (3.1)$$

The R-151 endpoint-loss upper envelope is

$$L(x) = \frac{624(7083/2000)x}{4096f(x)f(4x)}. \quad (3.2)$$

For the target 19/25, clear the positive denominator. The integer-scaled quartic is

$$\begin{aligned} \hat{P}_{19}(x) &= 30400000000000000000000x^4 - 35160465678800000000000x^3 \\ &\quad + 22722309685421961814576x^2 - 5761747682936108571810x \\ &\quad + 478871787740547514851. \end{aligned} \quad (3.3)$$

Its exact Sturm chain has degrees $(4, 3, 2, 1, 0)$. The signs at zero and positive infinity are respectively

$$(+, -, -, +, +), \quad (+, +, -, -, +), \quad (3.4)$$

both with two variations. Since the constant term is positive, there is no positive root and $L(x) < 19/25$ for every $x \geq 0$.

The exact minimum

$$\min_{y \geq 0} f(y) = \frac{28800000000947494031}{10^{20}} > \frac{36}{125} \quad (3.5)$$

gives $1/\lambda_2 < 125/36$. With $u = \sqrt{x}$, the second integer quartic

$$160000000000u^4 - 37011016504u^2 - 75000000000u + 5020336473 \quad (3.6)$$

represents $10^{10}\{f(4u^2) - 3u/4\}$. Its Sturm signs are

$$(+, -, -, +, +), \quad (+, +, +, -, +), \quad (3.7)$$

again with two variations. Hence

$$\frac{|p|}{\lambda_2} < \frac{4}{3}. \quad (3.8)$$

Using $P > 4$ in (2.11) gives the momentum-uniform sufficient collar

$$C_{\text{past}} < \frac{1524\sqrt{3}}{125}NM_W + \frac{787\sqrt{3}}{16}NM_Y + \frac{787}{32}N^2. \quad (3.9)$$

4. Nonlinear predictable directions

Take a legal fixed-law control direction

$$h_2(t, \xi_1) = t\phi(\xi_1), \quad z = S_2\phi, \quad y_i = D_{2,i}\phi. \quad (4.1)$$

The exact endpoint Hessian is

$$\begin{aligned} \mathcal{R}''(0)[\phi, \phi] = \int \mathbb{E} \left[\sum_i \left\{ y_i^* B y_i + 2y_i^* D B[z] V_i + \frac{1}{2} V_i^* D^2 B[z, z] V_i \right\} \right. \\ \left. - \frac{1}{2} \text{Tr}\{D^2 B[z, z] Q_0\} \right] dx. \end{aligned} \quad (4.2)$$

There is no $D_{\xi_1}\phi$ or feedback-connection term in (4.2): ϕ is held fixed as an element of the predictable control space.

Let $K(\xi_1)$ be the measurable symmetric twelve-real conditional Hessian of the complete endpoint plus terminal sixth-power term, excluding the source cost, in the ξ_2 source metric. On its bounded predictable simple form core,

$$\mathcal{A}''(0)[\phi, \phi] = \mathbb{E} \left[\phi^* \left(\frac{9}{10} \text{I} + K(\xi_1) \right) \phi \right]. \quad (4.3)$$

Theorem 4.1 (exact conditional-operator gate)

Subject to the displayed form-domain integrability, the following are equivalent:

$$\mathcal{A}''(0)[\phi, \phi] \geq \frac{1}{10} \mathbb{E}|\phi|^2 \quad \text{for every bounded predictable simple } \phi, \quad (4.4)$$

$$\boxed{K(\xi_1) \succeq -\frac{4}{5} \text{I} \quad \text{almost surely.}} \quad (4.5)$$

Sufficiency is immediate. For necessity, test rational directions q on measurable sets where $q^*(K+4I/5)q < -\varepsilon$ and truncate to $\|K\| \leq M$. A stronger uniform strict gap requires $K \succeq (-4/5 + \delta)I$. Extension from the simple core to arbitrary L^2 directions requires a declared closed form domain.

R-151 tests only the finite family $\phi = H\xi_1$ in averaged expectation. It does not prove (4.5). The logical gap is exact. Let $X \sim N(0, 1)$ and on the natural form domain take

$$K(X) = -cX^2, \quad c = \frac{1}{5}. \quad (4.6)$$

For every linear $\phi = HX$, the normalized adverse coefficient is $3c = 3/5 < 4/5$. Let $\chi \in C_c^\infty(-1, 1)$ be nonzero and $\phi_R(x) = \chi(x - R)$. Its ordinary C^3 norms are independent of R , while

$$\frac{\mathbb{E}[K(X)\phi_R(X)^2]}{\mathbb{E}[\phi_R(X)^2]} \leq -c(R-1)^2. \quad (4.7)$$

At $R = 4$ the source-plus-owner coefficient is at most $9/10 - 9/5 = -9/10$. Thus linear testing and ordinary bounded smoothness do not imply the conditional gate. This is a one-Gaussian-coordinate logical non-implication fixture on an unbounded multiplication-form domain, not the production operator K and not an A1 counterexample.

Two other derivatives must remain separate. A curved control parametrization Θ adds

$$D^2(\mathcal{A} \circ \Theta)[u, u] = D^2\mathcal{A}[D\Theta u, D\Theta u] + D\mathcal{A}[D^2\Theta(u, u)], \quad (4.8)$$

while Gaussian-source differentiation adds the R-128 inverse-Jacobian and $D\mathcal{A} \cdot D_\xi^2 h$ connection. Neither term belongs to (4.2), and neither vanishes without a full criticality theorem.

5. Multi-root one-use criterion

At the finite cylindrical/root-coordinate matrix level, let $\mathcal{X} = \bigoplus_e \mathcal{X}_e$ be the direct sum of local pair coordinates and let their actual predictable source incidence be

$$J = [J_e] : \mathcal{X} \longrightarrow \mathcal{S}, \quad G = J^*J. \quad (5.1)$$

The source Hessian is $9G/10$, once. If E, S, L denote the fully recombined endpoint, sixth-power, and genuine low Hessians, put

$$M = E + \frac{9}{10}G + S + L. \quad (5.2)$$

The exact global target is

$$\boxed{M \succeq \mu G \text{ for some } \mu > 0.} \quad (5.3)$$

For $G \succ 0$, this is the generalized-eigenvalue test

$$\lambda_{\min}(G^{-1/2}MG^{-1/2}) \geq \mu. \quad (5.4)$$

If G is singular, decompose $\mathcal{X} = \ker J \oplus (\ker J)^\perp$. The exact kernel-Schur conditions are

$$M_{00} \succeq 0, \quad \text{Ran } M_{01} \subseteq \text{Ran } M_{00}^{1/2}, \quad (5.5)$$

$$M_{11} - M_{10}M_{00}^\dagger M_{01} \succeq \mu G_{11}. \quad (5.6)$$

An infinite-dimensional predictable-control extension additionally requires a declared closed form domain, closed-range compatibility, and a well-defined Moore–Penrose form; fixed Fourier cutoff alone does not supply those facts.

An attempted local source allocation with fractions $0 \leq \theta_e \leq 1$ is legal only if

$$G - \text{diag}(\theta_e J_e^* J_e) \succeq 0. \quad (5.7)$$

At $\theta_e = 1$, the residual has zero diagonal blocks; positivity forces all off-diagonal $J_e^* J_f$ to vanish. Thus full local source payments can be summed only for genuinely orthogonal source incidences. Canonical Doob coordinates can diagonalize the source norm, but they do not delete the cross-root blocks of E, S, L .

Even source orthogonality and the strengthened local gap do not imply (5.3). Take $G = I_2$, $S = L = 0$, and

$$E = \begin{pmatrix} -3/4 & -1/5 \\ -1/5 & -3/4 \end{pmatrix}. \quad (5.8)$$

Each diagonal local action has gap $-3/4 + 9/10 = 3/20 > 7/50$, but

$$M = \begin{pmatrix} 3/20 & -1/5 \\ -1/5 & 3/20 \end{pmatrix} \quad \text{has eigenvalues} \quad -\frac{1}{20}, \frac{7}{20}. \quad (5.9)$$

This is an exact linear-algebra non-implication fixture. It is not the production cross-root matrix. Its consequence is that the actual production M must be constructed before any pairwise R-151/R-152 gaps are aggregated.

6. Route decision and evidence map

Route	Verdict	Reason and reuse boundary
Arbitrary field mean, zero current mean	advanced	Global R-130 envelopes preserve the core; exact loss is below 19/25.
Nonzero past current without a collar	inconclusive	The exact signed Δ_n is not paid by any registered once-owned estimate.
Linear tests imply nonlinear predictable gap	failed	The conditional operator gate is stronger; translated bumps give an exact non-implication.
Add local pair gaps after source orthogonalization	failed	Cross-root endpoint/sextic/low blocks survive; the two-edge fixture is globally negative.
Construct actual production M and test (5.3)	open	This is the only direct A13 globalization successor justified by the present analysis.

The bounded successor requested after R-151 has therefore reached its stop condition: no registered hypothesis supplies the past-current collar, the nonlinear conditional operator, or the full multi-root cross matrix. Further abstract local pair lemmas would not answer those missing production data. The mathematically honest action is to preserve R-151/R-152 at their scopes and move primary physical-model selection to the phase-neutral T-054 evidence register and candidate tournament. T-052 remains only the current-functional branch after the common comparison rules are frozen.

7. Devil's-advocate review

- Objection: arbitrary m means the R-151 proof is unchanged in every respect. VALID WITH MITIGATION.** The Hessian envelope is unchanged when $n = 0$, but stationarity and zero first variation are not.
- Objection: (2.13) proves the physical nonzero-past theorem. UPHELD against that inference.** The collar is a new conditional hypothesis and is not derived from the registered one-use budgets.

3. **Objection: nonlinear ϕ adds $D_\xi\phi$ to the fixed-law Hessian. DISMISSED.** It does not. Such derivatives belong to a Gaussian-source pullback or curved parametrization and carry their own connection terms.
4. **Objection: the bump fixture is a production counterexample. UPHELD against that inference.** It proves only that averaged linear tests and ordinary C^3 control do not imply an almost-sure conditional form bound.
5. **Objection: canonical Doob coordinates make all root blocks orthogonal. UPHELD.** They make the source norm one-use; they do not diagonalize the complete endpoint, sixth-power, or low Hessian.
6. **Objection: the negative eigenvalue in (5.9) falsifies TECT. UPHELD against that inference.** Equation (5.9) is a logical counterfixture, not the uncomputed production matrix.
7. **Objection: this result selects a phase or rejects the current PDE. UPHELD against that inference.** No phase ansatz, reference-state comparison, or candidate-functional discriminator occurs here.

External review is invited for the derivative factors in (2.4), the $\sqrt{3}$ Gaussian moment factor, both Sturm chains, the source metric in (4.3), measurable-form necessity in Theorem 4.1, and the singular- G Schur condition.

8. Result footer and reproduction

Result ID: A13-CLASSII-AFFINE-PAST-NONLINEAR-MULTIROOT-GLOBALIZATION-BOUNDARY (R-152).
Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.
Precise statement: The R-151 endpoint conditioned on a fixed strict past has the exact Hessian core plus Δ_n . Arbitrary field mean with zero current mean preserves endpoint loss below 19/25 and augmented curvature above 7/50. The declared $C_{\text{past}} \leq 1/25$ collar preserves curvature above 1/10. Every nonlinear predictable direction is controlled exactly by the complete conditional operator $K \geq -4I/5$ on its form core; R-151 linear tests do not imply this. Multi-root globalization requires the fully recombined $M \geq \mu G$ test; pairwise gaps do not imply it.
Scope: fixed finite cutoff and positive floor; retained admissible antipodal p:2p chart; fixed conditional affine past; bounded predictable simple nonlinear form core; abstract two-edge aggregation audit.
Dependencies: A1; R-119, R-125, R-128, R-130, R-133, R-140, R-141, R-151.
Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe codes/foundations/a13_classii_affine_past_nonlinear_multiroot_globalization_boundary_verify.py
Expected output: R-152 integrated PASS; primary 32/32 and non-importing standard-library independent 26/26 children PASS; PDF and generated-surface gates PASS.
Falsification gate: failure of the endpoint chain rule; either Sturm chain; the current-collar constants; conditional threshold; smooth-bump non-implication; source-allocation determinant; two-edge eigenvalues; any pinned authority, executable, JSON, exploration, negative, PDF, manifest, or public-surface assertion.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: no unconditional nonzero-past or nonlinear production gap, complete multi-root Hessian, T-050, A13, Nelson theorem, interacting measure, phase/PDE verdict, or Sector-A closure is proved.
Proof complete: false. Sector A closed: false.
Next required action: preserve A13 as conditional mathematics. Reopen its globalization route only with the actual production M or a derived past-current/conditional-operator theorem. Move primary physical-model work to T-054, freeze common evidence and candidate manifests, then run T-052 as the M1 branch and T-053 only on surviving candidates.

No-overclaim footer. R-152 closes the promised bounded post-R-151 review, not T-050. It proves one conditional affine-past curvature theorem and identifies two exact missing global operators. It does not establish the microscopic law of nature or any phase, vacuum, PDE, Nelson, interacting-measure, or Sector-A conclusion.